

Rajeev Jain

CONTACT Website: rajeeja.github.io GitHub: [rajeeja](#) LinkedIn: [rajeeja](#) Scholar:
HL=EN##
GOOGLE
SCHOLAR

jain@anl.gov U.S. Permanent Resident

Principal Software Engineer | ML Infrastructure | HPC | Scientific Computing

SUMMARY

Principal software engineer with 16+ years delivering production systems at the intersection of scientific computing, ML infrastructure, and high-performance computing. Led UXarray from zero to the standard Python tool for DOE labs and NCAR — including an MCP server that lets AI agents inspect, analyze, and visualize production Earth-system meshes on HPC clusters via Globus Compute with full provenance. Built the CANDLE/Supervisor HPO workflow (Swift/T, mlrMBO, DEAP) to run 10,000+ cancer AI experiments on Summit and Theta; applied noise injection and counterfactual analysis to identify gene-level decision drivers. Reduced FLASH-X checkpoint overhead 40–70% via async HDF5 and lossy compression. Ported Pangu-Weather to 60,000+ Intel XPU on Aurora. Two R&D 100 Awards (2022, 2023). Equally comfortable leading open-source Python platforms, C++ and Fortran HPC codebases, exascale GPU porting, and cross-institution program coordination.

PROFESSIONAL EXPERIENCE

Argonne National Laboratory

2009 – Present

Principal Specialist, Research Software Engineering Mathematics and Computer Science Division, Chicago, IL

- **UXarray and MCP server.** Lead developer for an open-source Python library for unstructured climate grid analysis (NCAR, DOE labs, universities). Designed the core API, conservative zonal averaging, grid I/O for ESMF / MPAS / SCRIP / HEALPix, CI/CD, and release workflows. Built the UXarray MCP server: 20+ typed tools that let AI agents inspect, subset, visualize, and run analysis on production Earth-system meshes locally or on HPC clusters (Argonne Improv, UCAR Derecho) via Globus Compute, with structured provenance on every operation. PBS-backed validation campaign on facility filesystems; companion paper submitted to SciFM 2026.
- **Pangu-Weather on Aurora.** Ported the Pangu-Weather / SFNO climate emulator to 60,000+ Intel XPUs on Aurora: PMIX/PALS launcher mapping, XPU/CUDA device branching, device-aware AMP (gradient scaling on CUDA, bf16 on XPU), and first stable DDP baseline. Established smoke-test and queue-aware bring-up scripts separating launcher, backend, and device-placement failure modes.
- **CANDLE / IMPROVE.** Built the CANDLE/Supervisor HPO workflow using Swift/T, mlrMBO (Bayesian/GP surrogate), DEAP (genetic algorithms), and Hyperopt/TPE; ran 10,000+ concurrent model training experiments across Summit, Theta, and Cori. Applied noise injection and counterfactual analysis to the NT3 tumor classification benchmark; identified PLOD2, RGS5, TP53I13, and related genes as decision-boundary drivers. Led the IMPROVE benchmark framework: `improvelib` Python package with standardized data loaders, canonical train/test splits, and evaluation metrics; GitHub Actions CI/CD covering unit tests, end-to-end Docker pipelines, and parallel cross-study analysis (CSA) via Parsl. Designed and implemented the UNO dual-branch neural network (separate drug and cell-line embeddings) for IMPROVE CSA benchmarking. Supported 15+ researcher multi-lab collaboration across Argonne, LLNL, and ORNL. R&D 100 Award, 2023.
- **FLASH-X.** I/O and compression lead for a million-line multiphysics simulation engine. Implemented asynchronous HDF5 I/O with Argobots and integrated SZ3 / ZFP compression, reducing checkpoint overhead 40–70% on Summit with 50%+ storage savings. Enabled cross-checkpoint restart between AMReX and Paramesh solvers. R&D 100 Award, 2022.
- **MeshKit and Urban ECP.** Principal investigator for MeshKit (DOE NEAMS, 2009–2016): open-source C++ toolkit for automated reactor core mesh generation. Developed parallel CoreGen: 712 processors, 101M hexahedral elements, 14 GB MONJU reactor mesh in under 7 minutes. Led urban exascale seed proposal with Charlie Catlett; built three-layer coupled workflow (WRF/HRRR → EnergyPlus → Nek5000 LES) and the full 3D downtown Chicago CFD geometry domain. Published in *Journal of Building Performance Simulation* (2020).
- Delivered release engineering, CI/CD, documentation, performance profiling, and reproducible workflows across climate, AI, and simulation software. Mentored students and research aides on scientific Python, HPC systems, and open-source engineering practices.

University of Chicago

Staff At-Large

2023 – Present

Chicago, IL

- Joint appointment supporting cancer pharmacogenomics and Earth science software collaborations.
- Mentor graduate students and research staff on software design, experiment reproducibility, and HPC workflows.

Arizona State University

2007 – 2009

Research and Teaching Assistant

Structural and Computational Mechanics Lab, Tempe, AZ

- FEM-based research on blast mitigation and design optimization; supported structural engineering coursework.

Earlier Experience

2005 – 2007

Project Engineer, Wipro Technologies | Engineering Intern, Tata Motors

India

TECHNICAL SKILLS

Languages: Python, C++, Fortran, Bash, R, SQL

ML / AI: PyTorch, TensorFlow, NumPy, Pandas, Xarray, Scikit-learn; Bayesian HPO (mlrMBO, Hyperopt/TPE), genetic algorithms (DEAP)

AI Agents / MCP: Model Context Protocol (MCP) server development, Claude API, LLM tool-use, Globus Compute for HPC-backed AI workflows

HPC / Systems: MPI, OpenMP, HDF5, NetCDF, MOAB, Argobots, SZ3/ZFP compression, Intel XPU / CUDA, PBS/SLURM job management

Workflow / DevOps: GitHub Actions, Docker, Singularity, Parsl, Swift/T, release automation, CI/CD, Globus Compute

Domains: Climate and weather modeling, cancer pharmacogenomics, computational physics, reactor mesh generation, urban microclimate

SELECTED OPEN SOURCE AND PROGRAM IMPACT

- **UXarray.** Lead developer; standard Python tool for DOE labs and NCAR for unstructured climate grid analysis. Tutorials and presentations at SC24, AMS 2024/2025, SciPy 2023, EGU, AGU, ESDS.
- **UXarray MCP Server.** AI-agent interface to UXarray: 20+ typed tools, Globus Compute HPC backend, structured provenance. Validated on Argonne Improv and UCAR Derecho. SciFM 2026 companion paper.
- **CANDLE / IMPROVE.** Core contributor to HPO workflow (Swift/T, mlrMBO, DEAP, Hyperopt) and improvelib benchmarking framework; GitHub Actions CI/CD; UNO dual-branch model. 15+ researcher multi-lab collaboration. R&D 100 Award, 2023.
- **FLASH-X.** I/O and compression lead for a million-line multiphysics engine (astrophysics, combustion, fluid dynamics). R&D 100 Award, 2022.
- **Pangu-Weather on Aurora.** Ported to 60,000+ Intel XPUs: PMIX/PALS launcher, XPU/CUDA AMP, first stable DDP baseline on Aurora.
- **MeshKit.** Principal investigator; open-source C++ reactor mesh generation toolkit; parallel CoreGen generated 101M-element MONJU mesh on 712 processors in under 7 minutes.

SELECTED PUBLICATIONS

- Partin, A., ..., **Jain, R.**, et al. “[Benchmarking community drug response prediction models.](#)” *Briefings in Bioinformatics*, 2025.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. “[Enabling Data Reduction for Flash-X Simulations.](#)” *SC24 / DRBSD-10*, 2024.
- **Jain, R.**, Wozniak, J. M., Partin, A., et al. “[Cross-HPO: Optimizing Neural Networks for Cancer Drug Response.](#)” *CAFCW at SC24*, 2024.
- **Jain, R.**, Luo, X., Sever, G., Hong, T., and Catlett, C. “Representation and evolution of urban weather boundary conditions in downtown Chicago.” *Journal of Building Performance Simulation*, 2020.
- Wozniak, J. M., **Jain, R.**, et al. “[CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research.](#)” *BMC Bioinformatics*, 2018.

Full list: [Google Scholar](#) (22+ publications).

AWARDS AND RECOGNITION

- R&D 100 Award, CANDLE (2023) | R&D 100 Award, FLASH-X (2022).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).

- Best Paper Award, International Meshing Roundtable (2010).

EDUCATION

University of Chicago	2020
M.S. in Computer Science	
Arizona State University	2009
M.S. in Structural Engineering	Graduate Fellowship Recipient
Indian Institute of Technology (ISM) Dhanbad	2006
B.Tech. in Mechanical Engineering	